

Calculus 1

Limits, Derivatives, and Integrals — A Working Cheat Sheet

mathcheatsheet

Abstract

A condensed but self-contained reference for single-variable calculus: limits and continuity, the derivative and its rules, applications of the derivative (monotonicity, optimization, related rates, linearization), the Riemann integral, the Fundamental Theorem of Calculus, techniques of integration, and applications of the integral. Each section pairs a tight statement of the theory with at least one worked example and a short list of common pitfalls.

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1 Limits

Definition 1.1 (ε - δ limit). $\lim_{x \rightarrow a} f(x) = L$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Intuition. “ $f(x)$ can be forced arbitrarily close to L by taking x sufficiently close to a , but $x \neq a$.” The value $f(a)$ is irrelevant — limits care about *approach*, not arrival.

1.1 Algebraic limit laws

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then

$$\begin{aligned} \lim(f \pm g) &= L \pm M, & \lim(fg) &= LM, \\ \lim \frac{f}{g} &= \frac{L}{M} \quad (M \neq 0), & \lim(cf) &= cL. \end{aligned}$$

For composition: if f is continuous at M and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} f(g(x)) = f(M)$.

1.2 One-sided and infinite limits

- $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$: approach from the left/right.
- Two-sided limit exists $\iff$ both one-sided limits exist and are equal.
- $\lim_{x \rightarrow a} f(x) = \infty$ means f grows without bound (no real L exists; this is a vertical asymptote).
- $\lim_{x \rightarrow \infty} f(x) = L$ describes horizontal asymptotes.

1.3 Standard limits worth memorizing

Must-know limits

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, & \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{1}{2}, & \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1, \\ \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} &= 1, & \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x &= e, & \lim_{x \rightarrow \infty} \frac{\ln x}{x^p} &= 0 \quad (p > 0).\end{aligned}$$

1.4 Squeeze theorem

Theorem 1.2 (Squeeze). Suppose $g(x) \leq f(x) \leq h(x)$ on a punctured neighborhood of a . If

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

then $\lim_{x \rightarrow a} f(x) = L$.

Example 1.3. $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$, because $-x^2 \leq x^2 \sin(1/x) \leq x^2$ and both bounds tend to 0.

1.5 Indeterminate forms and L'Hôpital

The forms $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0 are *indeterminate*: the limit depends on *how* the parts go to zero/infinity.

Theorem 1.4 (L'Hôpital's rule). If $\lim_{x \rightarrow a} \frac{f}{g}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, f, g are differentiable near a (with $g' \neq 0$), and $\lim_{x \rightarrow a} \frac{f'}{g'}$ exists, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Example 1.5. $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} \xrightarrow{\text{LH}} \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} \xrightarrow{\text{LH}} \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = -\frac{1}{6}.$

Pitfalls

- L'Hôpital only applies after verifying you have $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
- Cancel common factors first (e.g. $(x-1)$) — often there is no need for L'Hôpital.
- For 1^∞ , 0^0 , ∞^0 : take $\ln$, rewrite as $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then apply L'H.

2 Continuity

Definition 2.1 (Continuity at a point). f is continuous at a iff

$$\lim_{x \rightarrow a} f(x) = f(a).$$

This requires three things: $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and the two are equal.

2.1 Types of discontinuity

- **Removable:** $\lim_{x \rightarrow a} f(x)$ exists but is $\neq f(a)$ (or $f(a)$ is undefined). Plug the hole and you're done.
- **Jump:** left and right limits exist but differ. Step functions, sign function.
- **Infinite:** a one-sided limit is $\pm\infty$. Vertical asymptote.
- **Essential / oscillatory:** no limit exists; e.g. $\sin(1/x)$ at 0.

2.2 Continuity on an interval and standard theorems

A function is continuous on $[a, b]$ if it is continuous at every interior point and one-sided continuous at the endpoints. Polynomials, $\sin$, $\cos$, e^x , and rational functions on their domain are continuous everywhere they are defined.

Theorem 2.2 (Intermediate Value Theorem). *If f is continuous on $[a, b]$ and N lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ with $f(c) = N$.*

Theorem 2.3 (Extreme Value Theorem). *If f is continuous on $[a, b]$, then f attains a maximum and a minimum value on $[a, b]$.*

Example 2.4 (Root finding via IVT). Show $f(x) = x^3 - x - 1$ has a root in $[1, 2]$. Since f is a polynomial it is continuous, $f(1) = -1 < 0$ and $f(2) = 5 > 0$. By IVT there exists $c \in (1, 2)$ with $f(c) = 0$.

Heuristic

“Continuous on a closed bounded interval” is the hypothesis behind almost every existence theorem you will meet: IVT, EVT, Rolle, MVT. Lose either continuity or the closed bounded interval and the conclusions fail.

3 The Derivative

Definition 3.1 (Derivative).

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

when the limit exists. f is differentiable at a iff $f'(a)$ exists.

Intuition. $f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$, equivalently the instantaneous rate of change of f at a . The linear function $L(x) = f(a) + f'(a)(x - a)$ is the *best linear approximation* to f near a .

3.1 Differentiability and continuity

Proposition 3.2. *If f is differentiable at a , then f is continuous at a .*

The converse is false: $f(x) = |x|$ is continuous at 0 but not differentiable there (left slope -1 , right slope $+1$).

3.2 Notation

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{df}{dx}, \quad Df(x), \quad \dot{y} \text{ (physics, } t\text{-derivative)}.$$

Higher derivatives: f'' , $f^{(n)}$, $\frac{d^ny}{dx^n}$.

3.3 Geometric and physical meaning

- **Tangent line at $x = a$:** $y = f(a) + f'(a)(x - a)$.
- **Normal line at $x = a$:** $y = f(a) - \frac{1}{f'(a)}(x - a)$ (when $f'(a) \neq 0$).
- **Velocity / acceleration:** if $s(t)$ is position, then $v(t) = s'(t)$, $a(t) = s''(t)$.

Example 3.3 (Derivative from the definition). For $f(x) = x^2$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

Pitfalls

- Differentiability requires a *two-sided* limit of the difference quotient.
- Cusps and corners are continuous but *not* differentiable.
- Vertical tangents ($f' \rightarrow \pm\infty$) are also non-differentiable.

4 Differentiation Rules

4.1 Algebraic rules

For differentiable f, g and constant c :

$$\begin{aligned}(cf)' &= cf', & (f \pm g)' &= f' \pm g', \\(fg)' &= f'g + fg' & & \text{(product),} \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} & & \text{(quotient, } g \neq 0), \\(f \circ g)'(x) &= f'(g(x)) g'(x) & & \text{(chain rule).}\end{aligned}$$

4.2 Elementary derivatives

Table to memorize

$f(x)$	$f'(x)$
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arccos x$	$-1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$

4.3 Implicit differentiation

When y is defined implicitly by an equation $F(x, y) = 0$, differentiate both sides with respect to x , treating y as a function of x and applying the chain rule:

$$\frac{d}{dx}[y^n] = ny^{n-1} \cdot \frac{dy}{dx}.$$

Example 4.1 (Circle). Differentiate $x^2 + y^2 = 25$:

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$: slope $= -3/4$.

4.4 Logarithmic differentiation

Useful when f is a product/quotient/power of complicated factors, especially $f(x)^{g(x)}$. Take $\ln$ first:

$$\ln y = g(x) \ln f(x) \implies \frac{y'}{y} = g'(x) \ln f(x) + g(x) \frac{f'(x)}{f(x)}.$$

Example 4.2. $y = x^x \implies \ln y = x \ln x \implies \frac{y'}{y} = \ln x + 1 \implies y' = x^x (\ln x + 1)$.

4.5 Higher-order derivatives

$f^{(n)}(x)$ measures, loosely, how fast $f^{(n-1)}$ is changing. Common ones:

$$\frac{d^n}{dx^n} e^{ax} = a^n e^{ax}, \quad \frac{d^n}{dx^n} \sin x = \sin\left(x + \frac{n\pi}{2}\right).$$

5 Applications of the Derivative

5.1 Mean Value Theorem and Rolle's Theorem

Theorem 5.1 (Rolle). *If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Theorem 5.2 (Mean Value Theorem (MVT)). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Reading: the instantaneous rate matches the average rate somewhere on the interval. Corollary: if $f' \equiv 0$ on an interval, f is constant there.

5.2 Monotonicity and concavity

- $f' > 0$ on $I \implies f$ strictly increasing on I .
- $f' < 0$ on $I \implies f$ strictly decreasing on I .
- $f'' > 0 \implies f$ concave up (graph above tangent lines).
- $f'' < 0 \implies f$ concave down.
- Inflection point: f'' changes sign.

5.3 Local extrema

A *critical point* is a point where $f'(c) = 0$ or $f'(c)$ does not exist. Local extrema of a differentiable function occur only at critical points (Fermat).

First / Second derivative tests

First derivative test (at a critical point c):

f' changes	conclusion at c
$+$ $\rightarrow$ $-$	local max
$-$ $\rightarrow$ $+$	local min
no sign change	neither (inflection or saddle)

Second derivative test: if $f'(c) = 0$ and $f''(c) > 0$ then local min; if $f''(c) < 0$ then local max; if $f''(c) = 0$ the test is inconclusive.

5.4 Absolute extrema on $[a, b]$ (closed-interval method)

1. Find critical points of f in (a, b) .
2. Evaluate f at the critical points and at the endpoints a, b .
3. The largest of these is the absolute max; the smallest is the absolute min.

5.5 Optimization (word problems)

Recipe. (1) Identify the quantity to optimize Q ; (2) draw a picture and label variables; (3) write Q as a function of *one* variable using a constraint; (4) state the domain; (5) find critical points and test; (6) interpret in context.

Example 5.3 (Box of maximum volume). Cut squares of side x from a 20×20 sheet and fold up. Volume: $V(x) = x(20 - 2x)^2$ for $x \in [0, 10]$. Then $V'(x) = (20 - 2x)^2 - 4x(20 - 2x) = (20 - 2x)(20 - 6x)$. Critical points: $x = 10$ (endpoint, $V = 0$) and $x = 10/3$. Since $V(10/3) > 0$, that is the maximum.

5.6 Related rates

Recipe. Identify all changing quantities, write an equation relating them *at all times*, differentiate both sides with respect to t , plug in the instantaneous values.

Example 5.4 (Sliding ladder). A 10 ft ladder leans against a wall; bottom slides away at 1 ft/s. How fast does the top slide down when the bottom is 6 ft from the wall?

Let x = distance of foot, y = height of top. Then $x^2 + y^2 = 100$. Differentiate: $2xx' + 2yy' = 0$, so $y' = -\frac{x}{y}x'$. At $x = 6$, $y = 8$, $x' = 1$, hence $y' = -3/4$ ft/s (negative = descending).

5.7 Linear approximation and differentials

Near $x = a$:

$$f(x) \approx L(x) = f(a) + f'(a)(x - a), \quad dy = f'(x) dx.$$

Example 5.5. $\sqrt{4.04} \approx 2 + \frac{1}{2 \cdot 2} \cdot 0.04 = 2.01$ (true value ≈ 2.00998).

5.8 Newton's method

To solve $f(x) = 0$ iterate

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

i.e. replace f by its tangent and find that line's root. Converges quadratically near a simple root, provided $f'(x_*) \neq 0$ and x_0 is sufficiently close.

6 The Integral

6.1 Antiderivatives

Definition 6.1. F is an antiderivative of f on I if $F'(x) = f(x)$ for every $x \in I$. Two antiderivatives of the same function differ by a constant, so

$$\int f(x) dx = F(x) + C.$$

6.2 Riemann sums and the definite integral

Partition $[a, b]$ into n subintervals of width Δx_i and pick a sample point x_i^* in each. The Riemann sum is $\sum_{i=1}^n f(x_i^*)\Delta x_i$.

Definition 6.2 (Definite integral). If the Riemann sums converge to a single value as the maximum width tends to 0 (independent of the partition and the sample points), f is *Riemann integrable* on $[a, b]$ and we write

$$\int_a^b f(x) dx = \lim_{\max \Delta x_i \rightarrow 0} \sum_{i=1}^n f(x_i^*)\Delta x_i.$$

Continuous functions on $[a, b]$ are integrable; so are bounded functions with at most finitely many discontinuities.

6.3 Properties

$$\begin{aligned}\int_a^a f dx &= 0, & \int_a^b f dx &= -\int_b^a f dx, \\ \int_a^b (\alpha f + \beta g) dx &= \alpha \int_a^b f dx + \beta \int_a^b g dx, \\ \int_a^c f dx + \int_c^b f dx &= \int_a^b f dx, \\ f \leq g \text{ on } [a, b] &\implies \int_a^b f dx \leq \int_a^b g dx.\end{aligned}$$

6.4 Fundamental Theorem of Calculus

Theorem 6.3 (FTC, Part I). If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, then F is differentiable on (a, b) and $F'(x) = f(x)$.

Theorem 6.4 (FTC, Part II / Evaluation theorem). If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Intuition. Differentiation and integration are inverse operations: differentiate the integral and you get f back; integrate the derivative and you recover F up to its endpoint values.

6.5 Leibniz rule (variable limits)

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t) dt = f(b(x))b'(x) - f(a(x))a'(x).$$

6.6 Mean Value Theorem for Integrals

Theorem 6.5. If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ with

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The right-hand side is the *average value* of f on $[a, b]$.

6.7 Standard antiderivatives

Table to memorize

$f(x)$	$\int f(x) dx$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\sec x \tan x$	$\sec x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$
$\frac{1}{x\sqrt{x^2-1}}$	$\operatorname{arcsec} x + C$

7 Techniques of Integration

7.1 u -substitution

If $u = g(x)$ and $du = g'(x) dx$, then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Example 7.1. $\int 2x \cos(x^2) dx$: let $u = x^2$, $du = 2x dx$, so the integral is $\int \cos u du = \sin(u) + C = \sin(x^2) + C$.

For definite integrals, change the limits as well:

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

7.2 Integration by parts

From the product rule:

$$\int u dv = uv - \int v du.$$

LIATE heuristic for picking u (in priority order): **L**og, **I**nverse-trig, **A**lgebraic, **T**rig, **E**xponential.

Example 7.2. $\int x e^x dx$: $u = x$, $dv = e^x dx \Rightarrow du = dx$, $v = e^x$.

$$\int x e^x dx = x e^x - \int e^x dx = (x - 1)e^x + C.$$

Example 7.3 (Tabular for repeated parts). $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$ (differentiate x^2 thrice, integrate $\sin x$ thrice, alternate signs).

7.3 Trigonometric integrals

Powers of $\sin, \cos$:

- Odd power of $\sin$ or $\cos$: peel off one factor, convert the rest using $\sin^2 + \cos^2 = 1$, substitute.
- All even: use $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$.

Powers of $\sec, \tan$: use $\sec^2 = 1 + \tan^2$ and the derivative pair $(\sec x)' = \sec x \tan x$, $(\tan x)' = \sec^2 x$.

7.4 Trigonometric substitution

Integrand contains	Substitute	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 = \cos^2$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 = \sec^2$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 - 1 = \tan^2$

7.5 Partial fractions

For a proper rational function $P(x)/Q(x)$ (degree of $P <$ degree of Q), factor Q over $\mathbb{R}$ into linear and irreducible-quadratic factors and decompose:

$$\frac{P(x)}{(x-a)^k(x^2+bx+c)^m} = \sum_{i=1}^k \frac{A_i}{(x-a)^i} + \sum_{j=1}^m \frac{B_j x + C_j}{(x^2+bx+c)^j}.$$

Integrate term by term using $\ln|x-a|$ for linear and $\arctan / \ln(x^2+bx+c)$ for quadratic pieces.

Example 7.4. $\int \frac{1}{x^2-1} dx = \frac{1}{2} \int \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C.$

7.6 Improper integrals

- Infinite limits: $\int_a^\infty f dx = \lim_{b \rightarrow \infty} \int_a^b f dx$.
- Unbounded integrand at $c \in [a, b]$: split at c and take one-sided limits.
- Converges if the limit is finite; otherwise diverges.

p -test (the workhorse)

$$\int_1^\infty \frac{dx}{x^p} \text{ converges } \iff p > 1, \quad \int_0^1 \frac{dx}{x^p} \text{ converges } \iff p < 1.$$

Pitfalls

- Forgetting the $+C$ on indefinite integrals.
- In u -substitution: not converting the limits, or not back-substituting in the indefinite case.
- Trig substitution: forgetting to undo the substitution (write the answer in terms of x , not

θ).

- Partial fractions: degree of numerator must be *less than* denominator — do polynomial long division first if not.

8 Applications of the Integral

8.1 Area between curves

For continuous $f \geq g$ on $[a, b]$:

$$A = \int_a^b (f(x) - g(x)) dx.$$

If the curves cross, split the integral at the intersection points and take $|f - g|$ piecewise.

8.2 Volumes of revolution

Disks/washers (rotation about a horizontal axis $y = c$):

$$V = \pi \int_a^b \left((R(x) - c)^2 - (r(x) - c)^2 \right) dx,$$

where R, r are the outer/inner radii as functions of x .

Cylindrical shells (rotation about the y -axis):

$$V = 2\pi \int_a^b x |f(x)| dx.$$

Example 8.1 (Cone). Rotate $y = x$ for $x \in [0, h]$ about the x -axis (disks): $V = \pi \int_0^h x^2 dx = \pi h^3/3$.

8.3 Arc length

For $y = f(x)$ continuously differentiable on $[a, b]$:

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

8.4 Surface area of revolution

Rotation of $y = f(x) \geq 0$ about the x -axis:

$$S = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx.$$

8.5 Average value of a function

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx.$$

8.6 Physics applications

- **Displacement vs. distance:** $\Delta s = \int_a^b v(t) dt$ (signed); distance traveled $= \int_a^b |v(t)| dt$.
- **Work (variable force along a line):** $W = \int_a^b F(x) dx$.
- **Center of mass** of a thin rod of density $\rho(x)$ on $[a, b]$:

$$\bar{x} = \frac{\int_a^b x \rho(x) dx}{\int_a^b \rho(x) dx}.$$

8.7 Differential equations: separable

A separable ODE $\frac{dy}{dx} = g(x)h(y)$ is integrated by separating variables:

$$\int \frac{dy}{h(y)} = \int g(x) dx + C.$$

Example 8.2 (Newton's law of cooling). $T'(t) = -k(T - T_\infty) \Rightarrow \int \frac{dT}{T - T_\infty} = -k \int dt \Rightarrow T(t) = T_\infty + (T_0 - T_\infty)e^{-kt}$.

End-of-course mantra

Calc 1 = limits make derivatives precise, derivatives make integrals computable (via the FTC), and the two operations are inverses of one another. Everything else — optimization, related rates, volumes, arc length, ODEs — is a re-application of those two ideas to a geometric or physical setting.